

Chapter 2

Sets and Their Operations

Real-Life Motivation

Sets are everywhere: a class of students, fruits in a basket, days of the week. A **set** is a well-defined collection of objects called **elements**. Mastering sets is fundamental for algebra, probability, logic, functions, and combinatorics.

2.1 Definition of a Set

A **set** is a well-defined collection of distinct objects considered as a whole. The objects inside a set are called **elements**.

- If an object x is an element of set A , we write $x \in A$. - If x is not in A , we write $x \notin A$.

Example 1: Let A be the set of first four natural numbers:

$$A = \{1, 2, 3, 4\}, \quad 2 \in A, \quad 5 \notin A$$

Example 2: The set of vowels in English:

$$V = \{a, e, i, o, u\}$$

2.2 Methods of Representing a Set

2.2.1 Roster (Tabular) Method

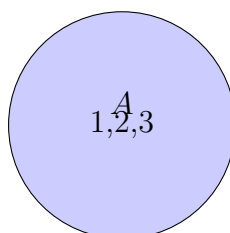
List all elements separated by commas inside curly braces. **Example:** $A = \{1, 2, 3, 4, 5\}$

2.2.2 Set-Builder Method

Describe the property that defines the elements. **Example:** $B = \{x \in \mathbb{N} \mid x \text{ is even}, x \leq 10\} = \{2, 4, 6, 8, 10\}$

2.2.3 Venn Diagram Representation

A set can also be represented as a circle containing all elements:



Types of Sets

Sets can be classified based on the number of elements, membership, or other properties. The main types of sets are:

1. Empty Set (Null Set)

Definition: A set that contains no elements is called an empty set or null set. It is denoted by $\emptyset$ or $\{\}$. **Example:**

$$A = \{x \in \mathbb{N} \mid x < 1\} = \emptyset$$

Here, there are no natural numbers less than 1, so A is empty.

2. Finite Set

Definition: A set which contains a definite number of elements is called a finite set.

Example:

$$B = \{2, 4, 6, 8, 10\}$$

B has exactly 5 elements, so it is finite.

3. Infinite Set

Definition: A set that has infinitely many elements is called an infinite set. **Example:**

$$\mathbb{N} = \{1, 2, 3, 4, \dots\}, \quad \mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

Both sets have no end, so they are infinite.

4. Singleton Set

Definition: A set containing exactly one element is called a singleton set. **Example:**

$$C = \{0\}, \quad D = \{\pi\}$$

C contains only 0; D contains only π .

5. Subset and Proper Subset

Definition: If every element of set A is also in set B , then A is a subset of B , written $A \subseteq B$. If $A \subseteq B$ and $A \neq B$, then A is a proper subset of B , written $A \subset B$.

Example:

$$A = \{1, 2\}, \quad B = \{1, 2, 3\}$$

$A \subseteq B$ and $A \subset B$ (proper subset).

6. Universal Set

Definition: The universal set is the set that contains all objects under consideration in a particular discussion. It is usually denoted by U . **Example:** If we are talking about integers from 1 to 10, the universal set is

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

7. Disjoint Sets

Definition: Two sets A and B are called disjoint if they have no elements in common, i.e., $A \cap B = \emptyset$. **Example:**

$$A = \{1, 2, 3\}, \quad B = \{4, 5, 6\}, \quad A \cap B = \emptyset$$

8. Equal Sets

Definition: Two sets A and B are said to be equal if every element of A is in B and every element of B is in A , denoted $A = B$. **Example:**

$$A = \{1, 2, 3\}, \quad B = \{3, 2, 1\} \quad \Rightarrow \quad A = B$$

9. Overlapping Sets

Definition: Two sets A and B are overlapping if they have at least one element in common. **Example:**

$$A = \{1, 2, 3\}, \quad B = \{3, 4, 5\}, \quad A \cap B = \{3\}$$

10. Power Set

Definition: The set of all subsets of a set A is called the power set of A , denoted by $P(A)$. **Example:**

$$A = \{1, 2\}, \quad P(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$$

11. Singleton and Doubleton Sets (Optional)

- Singleton: set with 1 element, e.g., $\{5\}$ - Doubleton: set with 2 elements, e.g., $\{3, 7\}$

2.4 Intervals of Real Numbers

Definition: An interval is a subset of real numbers lying between two endpoints.

- Open Interval $(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$, endpoints excluded. Example: $(2, 5)$
- Closed Interval $[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$, endpoints included. Example: $[2, 5]$
- Half-Open Intervals $[a, b)$ or $(a, b]$ include one endpoint. Example: $[2, 5)$

2.5 Venn Diagrams

Definition: A **Venn diagram** is a visual representation of sets using circles (or other shapes) to show the relationships between them. - Each circle represents a set. - The area of overlap between circles represents common elements (intersection). - The area outside a circle but within the universal set represents elements not in the set (complement).

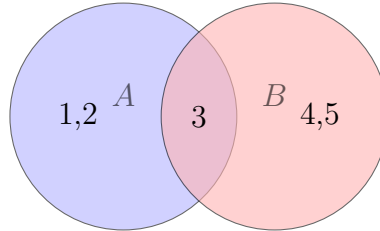
Purpose: Venn diagrams help to visually understand operations such as union, intersection, difference, and complement of sets.

Examples:

Example 1: Union and Intersection

Let $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$.

- Union: $A \cup B = \{1, 2, 3, 4, 5\}$ - Intersection: $A \cap B = \{3\}$

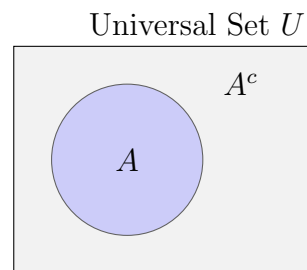


Here, the overlapping area represents $A \cap B = \{3\}$, and the combined area of both circles represents $A \cup B = \{1, 2, 3, 4, 5\}$.

Example 2: Complement

Let $U = \{1, 2, 3, 4, 5, 6\}$ and $A = \{2, 4, 6\}$. Then the complement of A is

$$A^c = \{x \in U \mid x \notin A\} = \{1, 3, 5\}$$



2.6 Operations on Sets

Definition: Operations on sets are rules that combine or modify sets to create new sets. The main operations are: **Union**, **Intersection**, **Difference**, **Complement**.

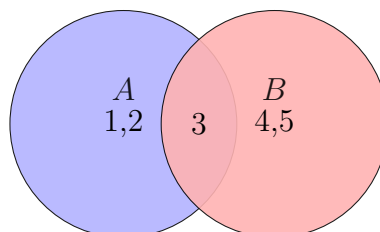
1. Union of Sets

Definition: The union of two sets A and B , denoted $A \cup B$, is the set of all elements which are in A , in B , or in both.

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

Example:

$$A = \{1, 2, 3\}, \quad B = \{3, 4, 5\} \quad \Rightarrow \quad A \cup B = \{1, 2, 3, 4, 5\}$$



2. Intersection of Sets

Definition: The intersection of two sets A and B , denoted $A \cap B$, is the set of all elements that are both in A and in B .

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

Example:

$$A = \{1, 2, 3\}, \quad B = \{3, 4, 5\} \quad \Rightarrow \quad A \cap B = \{3\}$$

3. Difference of Sets

Definition: The difference of sets A and B , denoted $A - B$, is the set of elements which are in A but not in B .

$$A - B = \{x \mid x \in A \text{ and } x \notin B\}$$

Example:

$$A = \{1, 2, 3\}, \quad B = \{3, 4, 5\} \quad \Rightarrow \quad A - B = \{1, 2\}$$

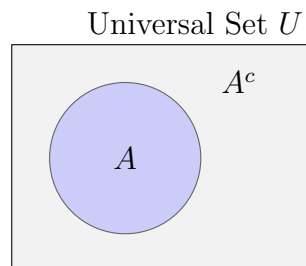
4. Complement of a Set

Definition: The complement of a set A with respect to a universal set U , denoted A^c , is the set of all elements in U that are not in A .

$$A^c = \{x \in U \mid x \notin A\}$$

Example:

$$U = \{1, 2, 3, 4, 5, 6\}, \quad A = \{2, 4, 6\} \quad \Rightarrow \quad A^c = \{1, 3, 5\}$$



Summary Notes

- $A \cup B$: Everything in A or B (or both)
- $A \cap B$: Only what is common to A and B
- $A - B$: Only elements in A not in B
- A^c : Everything in the universe not in A

2.9 Miscellaneous Solved Examples on Sets

Example 1: Determine which of the following sets are null sets: (i) The set of odd natural numbers divisible by 2. (ii) The set of even prime numbers greater than 2. (iii) The set $\{x \in \mathbb{N} : x < 5 \text{ and } x > 7\}$.

Solution:

- (i) No odd number is divisible by 2, so the set is empty.
- (ii) Only even prime is 2; no even prime greater than 2, so empty.
- (iii) No number satisfies $x < 5$ and $x > 7$ simultaneously, so empty.

Answer: $\emptyset$ for all three.

Example 2: Determine whether the following sets are finite or infinite: (i) The set of months in a year. (ii) The set of positive integers greater than 100. (iii) The set of prime numbers less than 99.

Solution:

- (i) 12 months $\Rightarrow$ finite.
- (ii) Integers > 100 continue indefinitely $\Rightarrow$ infinite.
- (iii) Finite number of primes $< 99 \Rightarrow$ finite.

Answer: (i) Finite, (ii) Infinite, (iii) Finite.

Example 3: Let $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4\}$. Verify if $A \subseteq B$.

Solution: All elements of A are in B . Therefore,

$$A \subseteq B$$

Answer: $A \subseteq B$.

Example 4: Find the union of $A = \{1, 3, 5\}$ and $B = \{2, 3, 4\}$.

Solution: Combine all elements without repetition:

$$A \cup B = \{1, 2, 3, 4, 5\}$$

Answer: $A \cup B = \{1, 2, 3, 4, 5\}$

Example 5: Find the intersection of $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$.

Solution:

$$A \cap B = \{3, 4\}$$

Answer: $A \cap B = \{3, 4\}$

Example 6: Find the difference $A - B$ and $B - A$ for $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$.

Solution:

$$A - B = \{1, 2\}, \quad B - A = \{5, 6\}$$

Answer: $A - B = \{1, 2\}, B - A = \{5, 6\}$

Example 7: Find the complement of $A = \{2, 4, 6\}$ in universal set $U = \{1, 2, 3, 4, 5, 6, 7\}$.

Solution:

$$A^c = \{1, 3, 5, 7\}$$

Answer: $A^c = \{1, 3, 5, 7\}$

Example 8: List all subsets of $X = \{a, b, c\}$ containing exactly two elements.

Solution:

$$\{a, b\}, \{a, c\}, \{b, c\}$$

Answer: $\{\{a, b\}, \{a, c\}, \{b, c\}\}$

Example 9: Let $A = \{1, 3\}, B = \{1, 5, 9\}, C = \{1, 3, 5, 7, 9\}$. Insert $\subset$ or $\in$ between each pair.

Solution:

(i) $\emptyset \subset B$

(ii) $A \cap B = \{1\} \subset C$

(iii) $A \subset C$

(iv) $B \subset C$

Answer: $\emptyset \subset B, A \cap B \subset C, A \subset C, B \subset C$

Example 10: Verify De Morgan's Law: $(A \cup B)^c = A^c \cap B^c$ for $A = \{1, 2\}, B = \{2, 3\}, U = \{1, 2, 3, 4\}$.

Solution:

$$\begin{aligned} A \cup B &= \{1, 2, 3\}, & (A \cup B)^c &= \{4\} \\ A^c &= \{3, 4\}, & B^c &= \{1, 4\}, & A^c \cap B^c &= \{4\} \end{aligned}$$

Answer: $(A \cup B)^c = A^c \cap B^c = \{4\}$

2.10 Unsolved Exercises on Sets

1. Determine whether the following sets are null sets: (a) The set of even numbers divisible by 3 and 5 but less than 2. (b) The set of prime numbers between 10 and 11. (c) The set of odd integers divisible by 4.
2. Let $A = \{1, 3, 5, 7\}$ and $B = \{2, 3, 4, 5\}$. Find $A \cup B, A \cap B, A - B, B - A$.
3. Find the complement of $A = \{2, 4, 6, 8\}$ in the universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$.
4. Let $X = \{a, b, c, d\}$ and $Y = \{b, d, f\}$. Verify if $X \subseteq Y, Y \subseteq X$.
5. Represent the following sets in roster form: (a) $\{x \in \mathbb{N} : x < 10, x \text{ even}\}$ (b) $\{x \in \mathbb{Z} : -3 \leq x \leq 3\}$
6. Write the following sets in set-builder form: (a) $\{1, 4, 9, 16, 25\}$ (b) $\{2, 4, 6, 8, 10\}$

7. Let $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, $C = \{1, 3, 5\}$. Find $(A \cup B) \cap C$ and $(A \cap C) \cup (B \cap C)$.
8. Draw a Venn diagram for $A = \{1, 2\}$, $B = \{2, 3\}$ and shade $A \cup B$, $A \cap B$, $A - B$.
9. Let $A = \{x \in \mathbb{N} : x \text{ is prime and } x < 10\}$ and $B = \{x \in \mathbb{N} : x \text{ is even and } x < 10\}$. Find $A \cup B$, $A \cap B$.
10. Find the power set of $X = \{a, b, c\}$.
11. Verify De Morgan's law: $(A \cap B)^c = A^c \cup B^c$ for $A = \{1, 2\}$, $B = \{2, 3\}$, $U = \{1, 2, 3, 4\}$.
12. If $A = \{1, 3, 5\}$, $B = \{2, 4, 6\}$, $C = \{1, 2, 3, 4\}$, find $(A \cup B) \cap C$ and $(A \cap C) \cup (B \cap C)$.
13. Determine whether the following statements are true or false: (a) $\emptyset \subseteq \{1, 2, 3\}$ (b) $\{1\} \subseteq \{1\}$ (c) $\{1, 2\} \subset \{1, 2\}$ (d) $\{2, 3\} \subset \{1, 2, 3\}$
14. Represent the following intervals as sets: (a) $x \in \mathbb{R}, 0 \leq x \leq 5$ (b) $x \in \mathbb{Z}, -2 \leq x \leq 2$
15. Let $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, $C = \{1, 3, 5\}$. Verify whether $(A - B) \cup (B - C) = (A \cup B) - C$.

Answer Key:

1. (a) $\emptyset$, (b) $\emptyset$, (c) $\emptyset$
2. $A \cup B = \{1, 2, 3, 4, 5, 7\}$, $A \cap B = \{3, 5\}$, $A - B = \{1, 7\}$, $B - A = \{2, 4\}$
3. $A^c = \{1, 3, 5, 7, 9\}$
4. $X \subseteq Y$: False, $Y \subseteq X$: False
5. (a) $\{2, 4, 6, 8\}$, (b) $\{-3, -2, -1, 0, 1, 2, 3\}$
6. (a) $\{x \in \mathbb{N} : x = n^2, 1 \leq n \leq 5\}$, (b) $\{x \in \mathbb{N} : x \text{ even}, 1 \leq x \leq 10\}$
7. $(A \cup B) \cap C = \{3\}$, $(A \cap C) \cup (B \cap C) = \{3\}$
8. Venn diagram: $A \cup B = \{1, 2, 3\}$, $A \cap B = \{2\}$, $A - B = \{1\}$ (draw circles with overlap)
9. $A \cup B = \{2, 3, 5, 7\}$, $A \cap B = \emptyset$
10. $\mathcal{P}(X) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$
11. $(A \cap B)^c = \{1, 3, 4\} = A^c \cup B^c$
12. $(A \cup B) \cap C = \{1, 2, 3, 4\}$, $(A \cap C) \cup (B \cap C) = \{1, 2, 3, 4\}$
13. (a) True, (b) True, (c) False, (d) True
14. (a) $\{x \in \mathbb{R} : 0 \leq x \leq 5\}$, (b) $\{-2, -1, 0, 1, 2\}$
15. $(A - B) \cup (B - C) = \{1, 2, 4, 6\} = (A \cup B) - C$